

AI Notes

1. Artificial Intelligence

Artificial Intelligence (AI) is the simulation of human intelligence by computers. AI systems can learn, reason, solve problems, and make decisions.

2. Machine Learning

Machine Learning is a subset of AI where computers learn patterns from data instead of being explicitly programmed.

3. Deep Learning

Deep Learning uses artificial neural networks with many layers to solve complex tasks such as image recognition and language understanding.

4. Neural Networks

Neural networks are inspired by the human brain and consist of interconnected nodes (neurons) arranged in layers.

5. Reinforcement Learning

Reinforcement Learning trains an agent to make decisions by rewarding good actions and penalizing poor ones.

6. Natural Language Processing

NLP enables computers to understand, interpret, and generate human language.

7. Computer Vision

Computer Vision allows computers to understand images and videos.

8. Generative AI

Generative AI creates new content such as text, images, audio, and code using trained models.

9. Applications

Applications include healthcare, education, finance, cybersecurity, autonomous vehicles, and recommendation systems.

10. Challenges

Challenges include bias, privacy, security, explainability, and ethical use of AI.